Coding & Solution

Maximum Marks: 3 Marks

Date	Team ID	Project Name
15 March 2026	Batch 2023-27 | ACOE	Rising Waters – ML-Based Flood Prediction System

Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

7-Step ML Implementation Pipeline

The Rising Waters ML pipeline follows the exact 7-step framework specified in the IBM Skill Wallet project scope.

Step	Activity	Key Actions	Output
1	Environment Setup	Install Python 3.11, Flask, Scikit-learn, XGBoost, Pandas, Joblib, Matplotlib	requirements.txt
2	Dataset Collection	Generate 5,000-row synthetic dataset with 6 meteorological features + binary target	flood_dataset.csv
3	Data Visualisation	EDA: distribution plots, box plots, pairplot, correlation heatmap (Matplotlib/Seaborn)	4 EDA charts
4	Data Preprocessing	Mean imputation for nulls, IQR outlier capping, StandardScaler fit-transform	scaler.save
5	Model Building	Train Decision Tree, Random Forest, KNN, XGBoost; evaluate on 80/20 train-test split	4 trained models
6	Best Model Selection	Compare accuracy: XGBoost wins at 96.55%; serialize with Joblib	floods.save
7	Web App + Deployment	Flask app with 6 HTML pages; Docker containerize; deploy on IBM Cloud Foundry	Live web app

Flask Application Routes

Method	Route	Description	Template
GET	/	Home Dashboard — model stats, prediction counts	home.html
GET	/predict	Input Form — 6 meteorological parameter fields	predict.html
POST	/predict	Process form → validate → scale → predict → redirect	—
GET	/result/flood	Flood Chance — red banner, 6 emergency action cards	result_flood.html
GET	/result/noflood	No Flood — green banner, monitoring recommendations	result_noflood.html
GET	/history	Prediction audit log from predictions_log.csv	history.html
GET	/about	Architecture, model info, team, use-case scenarios	about.html